

Equations and Inequalities Problem Solving

Lesson 7-7

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

'Twice a number is 18' → $2n = 18$ — the equation models the words

Model

To show a real-life situation with an equation or inequality.

$x + 5 = 12$ — exactly one answer: $x = 7$

Equation

A math sentence with an equal sign showing both sides are the same.

$x + 5 > 12$ — many answers: $x = 8, 9, 10, \dots$

Inequality

A math sentence that compares two sides with $<$, $>$, $\leq$, or $\geq$.

If the problem asks for a number of people and you get $x = -3$, that is NOT reasonable

Reasonableness

Checking if your answer makes sense.

Key Ideas & Notes

- Detective Santos is closing a complex case.
- She has collected equations and inequalities from different parts of the investigation.
- One clue says: 'The total value of stolen items divided equally among 4 accomplices gave each person \$85.' Another says: 'The lookout earned less than \$50.' She must model both clues mathematically and check if the answers are reasonable.
- Can you help?
- Read each word problem. Decide if it needs an equation or an inequality. Write the mathematical model, solve it, and check your answer for reasonableness.

Think About It

- Which clue leads to an equation and which leads to an inequality?
- What operation does 'divided equally among 4' suggest?
- What does 'less than \$50' tell you about the inequality symbol?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Before solving a word problem, how do you decide whether to model it with an equation or an inequality?

Level 1 support

Start here: An exact total points to an equation; a limit or range points to an inequality.

I would model this with a(n) _____ because the problem says _____.

Modelaría esto con un(a) _____ porque el problema dice _____.

Words like _____ tell me to use _____.

Palabras como _____ me dicen que use _____.

Word bank: **model** **equation** **inequality** **reasonableness**

Listen for: Students use signal words ('equals/total' for an equation; 'at least/at most/no more than' for an inequality) to choose the correct model.

Level 2

How would the same situation change if 'spends exactly \$50' became 'spends at most \$50'? Justify which model fits each version.

- 'Exactly \$50' needs a(n) _____, but 'at most \$50' needs a(n) _____ because _____.
- The model changes because _____.

EXPLORE

What was your plan to solve this problem, and how did you check that your answer was reasonable?

Level 1 support

Start here: I model the problem, solve it, then check the answer against the situation.

My plan was to ____, **and I knew to** ____ **because** ____.

Mi plan fue ____, *y supe* ____ *porque* ____.

My answer is reasonable because ____.

Mi respuesta es razonable porque ____.

Word bank: **model** **equation** **inequality** **reasonableness**

Listen for: Students describe modeling the situation, solving with inverse operations, and judging reasonableness by checking the answer fits the real context.

Level 2

Suppose your solution is a negative number or a fraction that doesn't fit the situation. What should you do, and why does reasonableness matter?

- If my answer doesn't fit, I should ____ because ____.
- Reasonableness matters because ____.

CONNECT

Describe a real situation that an equation or inequality could model, and explain which one you would use.

Level 1 support

Start here: Real situations with limits or exact totals can be modeled mathematically.

A real situation is _____, which I would model with a(n) _____.

Una situación real es _____, que modelaría con un(a) _____.

I chose this model because the situation has _____.

Elegí este modelo porque la situación tiene _____.

Word bank: **model** **equation** **inequality** **reasonableness**

Listen for: Students give a realistic context and correctly match it to an equation (exact value) or inequality (range/limit), justifying their choice.

Level 2

Critique: 'Any word problem can be solved with an equation, so inequalities are never needed.' Use an example to argue why inequalities are sometimes the only correct model.

- Inequalities are needed when _____.
- An example that cannot use a single equation is _____ because _____.

Guided Examples

Example 1

A museum has some paintings. After adding 12, they have 45. Which equation models this?

Solution: 'Adding 12' to the original number gives 45: $p + 12 = 45$. Solve: $p = 33$.

Answer: A. $p + 12 = 45$

Example 2

A detective needs more than 8 hours to finish the investigation. She has already worked 3 hours. Which inequality represents the additional hours h she needs?

Solution: 3 hours plus additional hours h must be more than 8: $3 + h > 8$. Solve: $h > 5$.

Answer: A. $3 + h > 8$

Example 3

Which model is correct for: 'A number divided by 6 equals 7'?

Solution: Divided by 6 is $n / 6$. Equals 7 means $= 7$. So $n / 6 = 7$.

Answer: A. $n / 6 = 7$

Write About the Math

The Writing Revolution

I can explain my reasoning using the words model, equation, inequality, and reasonableness.

1. Kernel Sentence subject + verb

Model: Equation is a math sentence with an equal sign showing both sides are the same.
Ecuación es una oración matemática con un signo igual que muestra que ambos lados son iguales.

Write a kernel sentence about equation. Use a subject and a verb.

Escribe una oración base sobre ecuación. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Equation matters in math
Ecuación importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Equation matters in math because ____.
Ecuación importa en matemáticas porque ____.

but
pero

Equation matters in math, but ____.
Ecuación importa en matemáticas, pero ____.

so
entonces

Equation matters in math, so ____.
Ecuación importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement
Afirmación

Tell one true fact about equation.
Di un hecho verdadero sobre equation.

Equation ____.

Question
Pregunta

Ask a question about equation.
Haz una pregunta sobre equation.

How does ____ ?

¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about equation.
Muestra entusiasmo sobre equation.

Wow, ____ !

¡Guau, ____ !

Command
Mandato

Tell a partner what to do with equation.
Dile a un compañero qué hacer con equation.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

I would model this with a(n) ____ because the problem says ____.

Modelaría esto con un(a) ____ porque el problema dice ____.

Words like ____ tell me to use ____.

Palabras como ____ me dicen que use ____.

My plan was to ____, and I knew to ____ because ____.

Mi plan fue ____, y supe ____ porque ____.

Try It

Solve on your own. Check the answer key when you are done.

1. A bus can carry at most 48 passengers. There are already 31 on board. Which inequality shows how many more can board?

- A. $31 + p \leq 48$
- B. $31 + p > 48$
- C. $31 + p = 48$
- D. $p - 31 \leq 48$

Show your work:

2. A student solved $4n = 52$ and got $n = 13$. Is the answer reasonable if n represents the number of notebooks in a box?

- A. Yes — 13 notebooks per box is reasonable
- B. No — 13 is too many notebooks
- C. No — the answer should be 48
- D. Yes — but only if n is a fraction

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Create two word problems about the same situation: one that requires an equation and one that requires an inequality. Solve both and explain why the models are different.

Sentence starter: Equation problem: _____. Model: _____. Solution: _____. Inequality problem: _____. Model: _____. Solution: _____. The models are different because _____.

Show your work:

Reflect — Exit Ticket

A box of donuts has some donuts. After giving away 7, there are fewer than 5 left. Which inequality represents the starting number of donuts d ?

- A. $d - 7 < 5$
- B. $d + 7 < 5$
- C. $d - 7 > 5$
- D. $d - 7 = 5$

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. $31 + p \leq 48$ — *'At most 48' means the total must be ≤ 48 . So $31 + p \leq 48$. Solve: $p \leq 17$.*
2. **Try It 2:** A. Yes — 13 notebooks per box is reasonable — $4 \times 13 = 52$ ✓. *13 notebooks per box is a reasonable whole number answer.*
3. **Exit Ticket:** A. $d - 7 < 5$ — *Starting with d donuts and giving away 7 leaves fewer than 5: $d - 7 < 5$. Solve: $d < 12$. ✓*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Equation is a math sentence with an equal sign showing both sides are the same.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).